

# Hariharen S S

☎ 9894545969   ✉ [thisishariharen@gmail.com](mailto:thisishariharen@gmail.com)   🌐 [hariharen.site](http://hariharen.site)   💼 [linkedin.com/hariharen9](https://linkedin.com/hariharen9)   🐙 [github.com/hariharen9](https://github.com/hariharen9)

*Software Engineer with 4+ years of experience designing, automating, and operating distributed microservices and cloud-native infrastructure at IBM Cloud. Proficient in Go, Python, Docker, and Kubernetes with a proven record of eliminating operational toil through controller-native solutions and automation-first engineering. Active open-source developer with nearly 20,000 downloads across published CLI and AI tools, alongside published research in cloud systems architecture.*

## EXPERIENCE

### Tata Consultancy Services

July 2022 – Present

*Cloud Software Engineer – Client: IBM Cloud*

*Bangalore, Karnataka*

- Built a parallelized Python CLI to automate Kubernetes network topology provisioning and CRD validation across multiple environments, replacing error-prone manual workflows with a deterministic orchestration system.
- Reduced end-to-end environment setup time by 95% (from hours to minutes) and drove cross-team adoption as the standard tooling for network CRD validation, integrated debugging, and teardown automation.
- Engineered a Go controller-native garbage collector to replace brittle legacy cleanup scripts, reducing orphaned Kubernetes object accumulation to near-zero and cutting infrastructure maintenance overhead by 90%.
- Led the production migration from Mezmo (LogDNA) to IBM Cloud Logs (ICL), standardizing logging pipelines across all environments, improving alert fidelity, and reducing observability operational costs.
- Elevated system reliability by defining SLI/SLO-aligned observability standards (Prometheus/Grafana dashboards, alerting runbooks, and incident playbooks) and resolving critical runtime panics and production bugs across key microservices.

### Halcyon

March 2020 – June 2022

*Co-Founder & Lead Developer*

*Coimbatore, Tamil Nadu*

- Co-founded a software development agency and delivered 5+ production-grade solutions for SMB clients, owning the full SDLC from requirements gathering to deployment across mobile, web, and backend stacks.
- Built a inventory and billing management system for a retail business and multiple Flutter-based cross-platform mobile applications, driving measurable efficiency gains and repeat client engagements.

## TECHNICAL SKILLS

**Languages:** Go, Python, TypeScript

**Backend & APIs:** gRPC, REST, Firebase, Supabase

**Cloud & Infrastructure:** IBM Cloud, AWS, Azure

**CI/CD:** Jenkins, GitHub Actions, Azure DevOps

**Containers & Orchestration:** Docker, Kubernetes, OpenShift

**Databases:** PostgreSQL, MongoDB, Firestore, SQL

**Frontend:** React, Next.js, Flutter

**Observability:** Prometheus, Grafana, IBM Cloud Logs

## CERTIFICATIONS & PUBLICATIONS

AWS Cloud Practitioner - **AWS**

IBM Cloud Advocate V2 - **IBM**

**Paper:** Deconstructing the Netflix Machine: A Case Study of Netflix's Production-to-Delivery Cloud Ecosystem – **SSRN**, Sep 2025

## PERSONAL PROJECTS

### CLI & Automation Tools

| *Go, Python*

- Built **FFmpeg-this**, an open-source Python CLI that translates human-readable media operation syntax into complex FFmpeg pipelines; published on PyPI with **7,000+ downloads**.
- Built **Kessler**, a **Go** TUI and VS Code extension that reclaims disk space by scanning **30+ language ecosystems** for build caches and dependency folders, using a custom rules engine that respects active processes and gitignore rules.
- Created **GitSync**, a Go CLI automating multi-branch synchronization, upstream rebasing, and stale branch cleanup — eliminating repetitive manual Git workflows for multi-remote repositories.

### AI Developer Tools

| *TypeScript, Python, Go*

- Created **Enhance-this**, a cross-platform CLI utility (npm/pip) for local AI prompt optimization, achieving **12,000+ organic downloads** by solving high-friction developer workflows.
- Engineered **Localseek**, a VS Code extension enabling fully offline AI code assistance via local models, achieving **500+ installs** and eliminating cloud API dependencies to preserve developer privacy.

### Automated Finance Tracker (SpendWiser)

| *React, Firebase, Netlify*

- Built a native Android bridge (BroadcastReceiver + React Native Headless JS) to intercept bank SMS alerts at the OS level, bypassing restrictive app store policies and eliminating dependency on expensive third-party sync APIs (e.g., Plaid).
- Architected a privacy-first serverless pipeline with on-device SMS pre-filtering and a cascading bring-your-own AI model engine that autonomously parses unstructured transaction data into a real-time personal finance dashboard.

## EDUCATION

### Coimbatore Institute of Technology

June 2019 – April 2022

*Bachelor of Engineering in Computer Science (CGPA – 8.7/10)*

*Coimbatore, Tamil Nadu*

### PSG College of Technology

June 2016 – April 2019

*Diploma in Electronics and Communication Engineering (CGPA – 9.5/10)*

*Coimbatore, Tamil Nadu*